

Homology modelling using MODELLER



Presented by: Bhumika Choudhary

PhD(CB) - MT22198

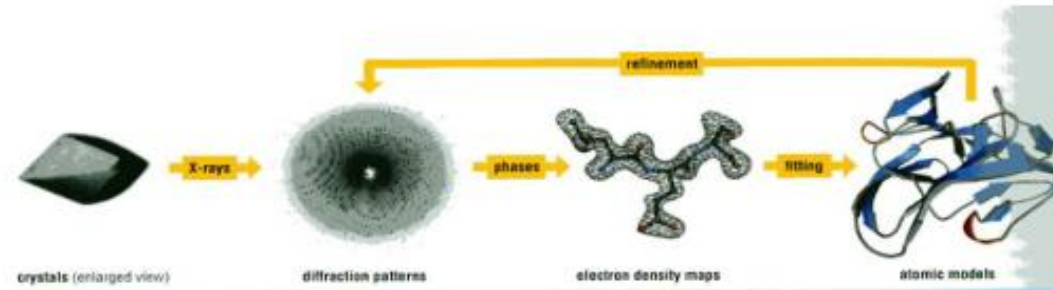
Instructor: Dr. Arjun Ray

Indraprastha Institute of Information Technology Delhi (IIIT-Delhi)

Protein Structure Determination

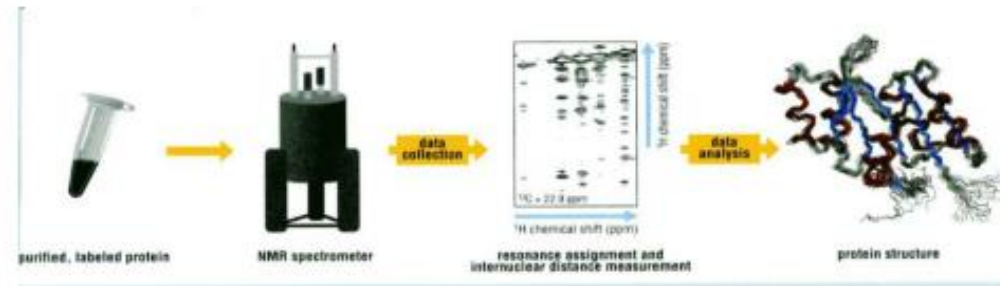
■ X-ray crystallography

- most accurate
- in vitro
- need crystals proteins
- ~50K per structure



■ NMR

- accurate
- in vivo
- no need for crystals
- limited to small proteins



■ Cryo-EM

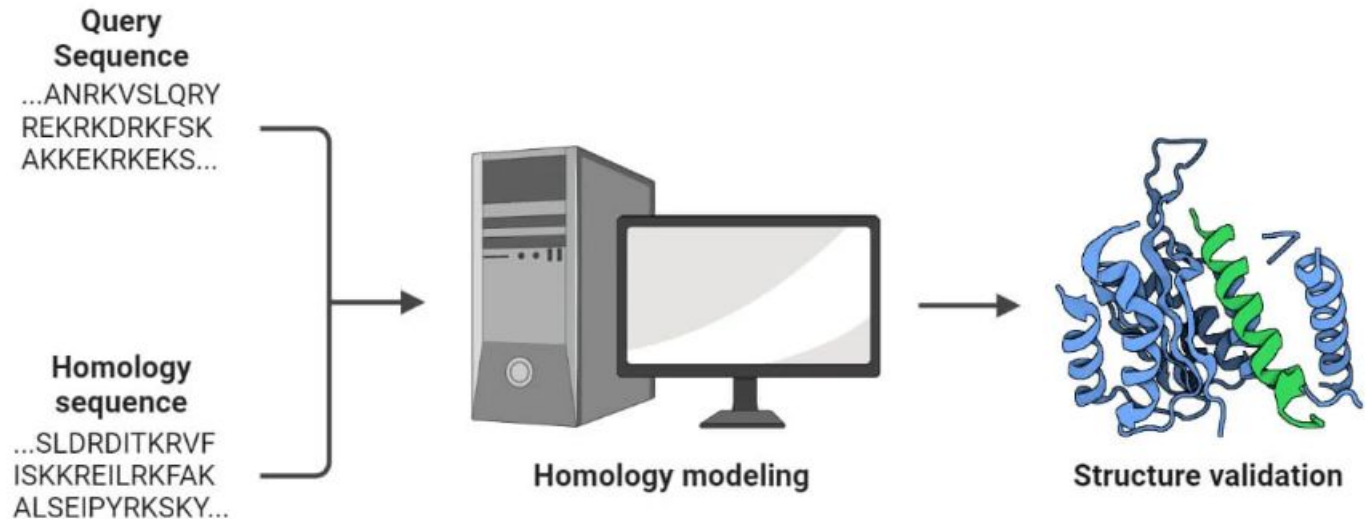
- Imaging technology
- Low-resolution

Homology Modelling: Introduction

- computational structure prediction method used to determine protein 3D structure from:
 - Its amino acid sequence
 - An experimented 3D structure of a related homologous protein “ **template**”
- method relies on the fact that the 3D structure of proteins is often better conserved than their amino acid sequence.
- Also known as **comparative modelling**

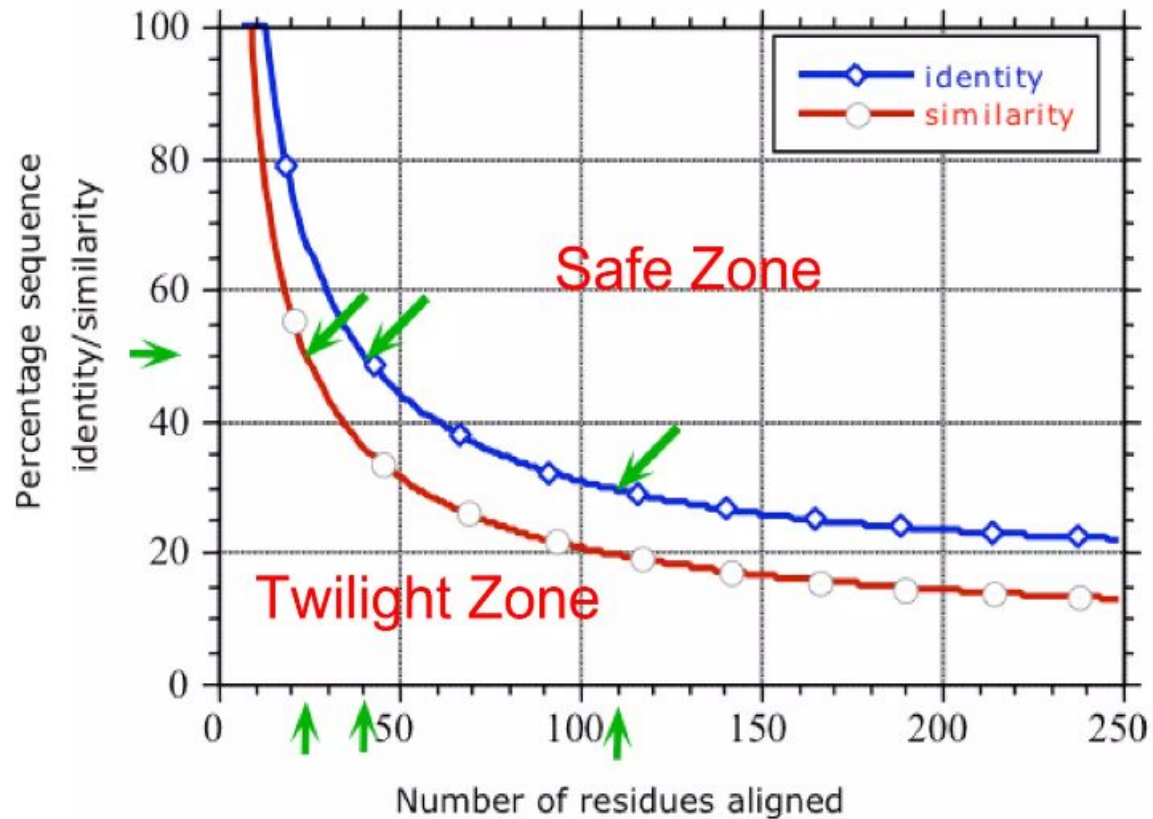
Homology modeling tools or servers

- **Modeller**
- i-tasser
- swiss-model
- Robetta
- Molecular Operating Environment (MOE)
- And many more....



Sequence similarity structural similarity

- High sequence similarity implies similar protein folds and structures, reflecting evolutionary relationships and shared function.
- Various algorithms and methods, such as **BLAST** are used to assess sequence similarity
- In general, **30% sequence similarity**, to generate a useful model.



NIH National Library of Medicine
National Center for Biotechnology Information

BLAST®

Home Recent Results Saved Strategies Help

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

Web BLAST

Nucleotide BLAST
nucleotide → nucleotide

blastx
translated nucleotide → protein

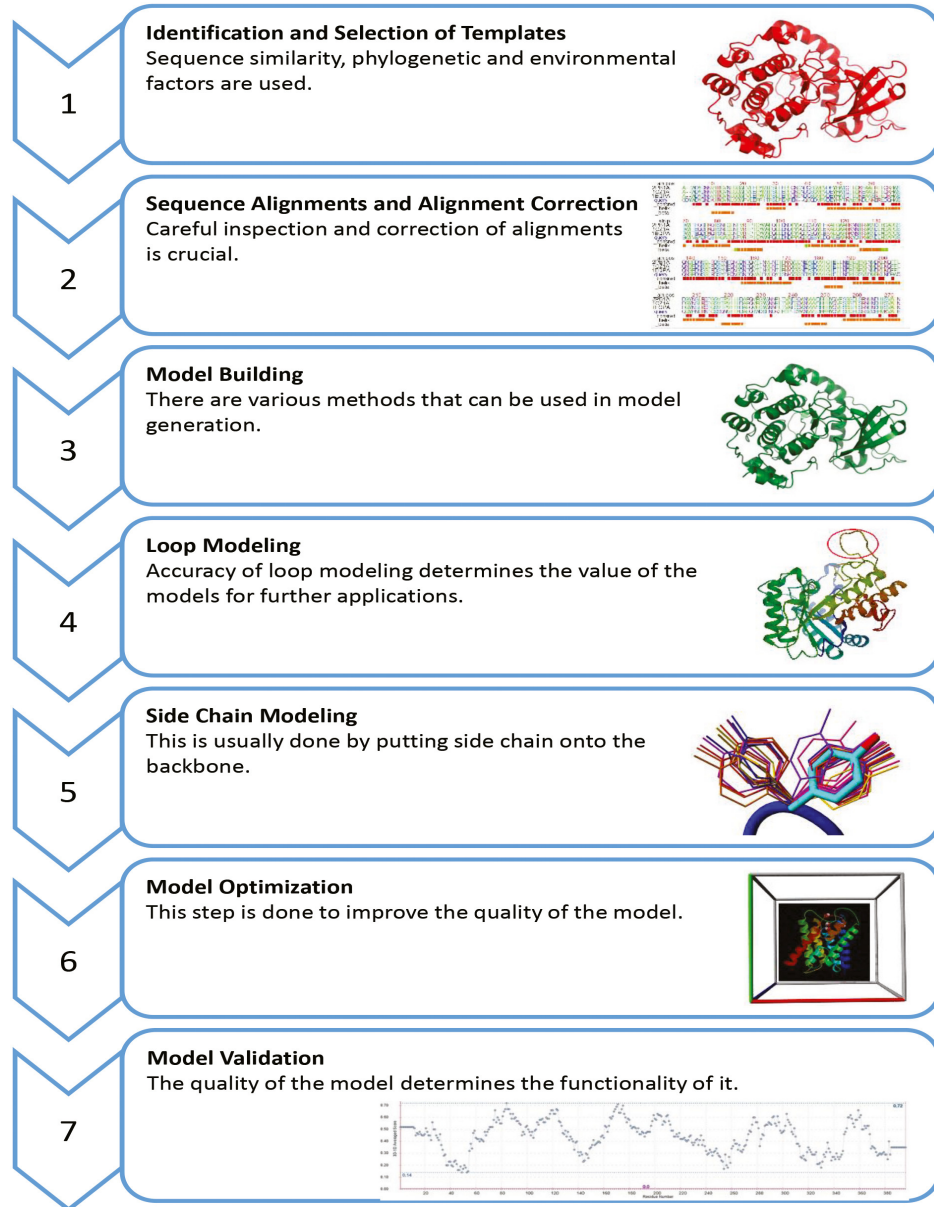
tblastn
protein → translated nucleotide

Protein BLAST
protein → protein

BLAST+ 2.15.0 is here!
We have included two exciting new features in the latest BLAST+ release
Tue, 28 Nov 2023 [More BLAST news...](#)

As long as the length of two sequences and the percentage of identical residues fall in the region marked as “safe” the two sequences are practically guaranteed to adopt a similar structure

Steps involved in homology modelling



Modeller

Program for Comparative Protein
Structure Modelling by Satisfaction
of Spatial Restraints



```
A I L V G S M P R R D G M E R K D L L K A N V K I F K C Q G A  
V E V C P V D C F Y E G P N F L V I H P D E C I D G A L C E P  
V A C K P E C P V N I I Q G S . . . Y A I D A D S C I D C G S  
Q . . A C G A C K P E C P V N I I Q G S . . . Y A I D A D S
```

- Freely available tool used for homology or comparative modeling of protein three-dimensional structures.
- ✓ **Basic modelling**
- ✓ **Advanced modelling**
- ✓ Iterative modelling
- ✓ Difficult modelling
- ✓ Modelling with cryoEM

Modelling Structure of Human c-Yes Tyrosine Kinase (Uniprot ID- P07947) (basic modeling)

Steps

- ☐ Searching and Selecting for template structure using BLAST
(Model only Kinase domain)
- ☐ Aligning the query with the template
- ☐ Build Kinase domain of c-Yes Kinase - Generate 5 models
- ☐ Model Evaluation – Calculate DOPE Score (lowest denotes best model)
- ☐ Report Structural similarity and Visualise the structures

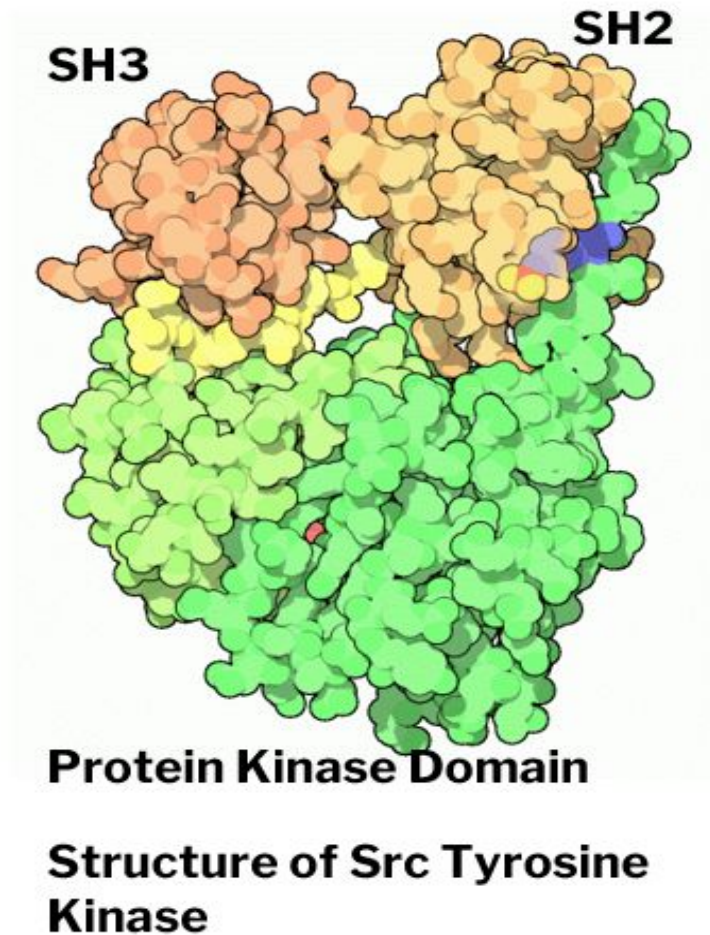
Pre-requisites

- ✓ Installation and registration of Modeller
- ✓ Tool for Visualisation, RMSD calculation, etc.

Modeller Installation (Conda –Add license key to config.py)

<C:/Users/91809/anaconda3/Library/modeller/modlib/modeller/config.py>

https://salilab.org/modeller/download_installation.html



Step 1: Download the query sequence and convert in PIR format.

'align code': an identifier used to identify the sequence. <PDBcode + chainID>

Sequence/Structure: 9 fields separated by colon
code:res_start:chain:res_end:chain:protein_name:source_protein:Resolution:R_factor

C; A sample alignment in the PIR format; used in tutorial

>P1;5fd1

structureX:5fd1:1 :A:106 :A:ferredoxin:Azotobacter vinelandii: 1.90: 0.19
AFVVTDCIKCKYTDCEVCPVDCFYEGPNFLVIHPDECIDCALCEPECQAIFSEDEVPEDMQEFIQLNAELA
EVWPNITEKKDPLPDAEDWDGVKGKQLQHLE*

>P1;1fdx

sequence:1fdx:1 : :54 : :ferredoxin:Peptococcus aerogenes: 2.00:-1.00
AYVINDSC--IACGACKPECPVNIIQGS--IY AIDADSCIDCGSCASVCPVGAPNPED-----
-----*

Sequence

Each sequence must be terminated by the terminating character, '*'.

Step 2: Selecting a template using BLAST

Sequences producing significant alignments

Download

Select columns

Show

100

?

☒ select all 100 sequences selected

[GenPept](#)

[Graphics](#)

[Distance tree of results](#)

[Multiple alignment](#)

[MSA Viewer](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Src in complex with DNA-templated macrocyclic inhibitor MC4b [Gallus gallus]	Gallus gallus	481	481	99%	5e-174	89.76%	275	3U4W A
☒	Crystal Structure of chicken c-Src kinase domain in complex with the cancer drug imatinib [Gallus gallus]	Gallus gallus	481	481	99%	7e-174	89.76%	286	2OIQ A
☒	Structural basis for the recognition of c-Src by its inactivator Csk [Gallus gallus]	Gallus gallus	481	481	99%	7e-174	89.76%	277	3D7U B
☒	c-Src kinase domain in complex with Rao-IV-151 [Gallus gallus]	Gallus gallus	479	479	99%	3e-173	89.76%	286	5T0P A
☒	human Src kinase bound to kinase inhibitor bosutinib [Homo sapiens]	Homo sapiens	481	481	99%	6e-174	89.37%	286	4MXO A
☒	Src kinase in complex with inhibitor AP23451 [Homo sapiens]	Homo sapiens	481	481	99%	8e-174	89.37%	279	2BDF A
☒	Chain A, Proto-oncogene tyrosine-protein kinase Src [Homo sapiens]	Homo sapiens	481	481	99%	8e-174	89.37%	287	7NG7 A
☒	Chain A, Proto-oncogene tyrosine-protein kinase Src [Homo sapiens]	Homo sapiens	481	481	99%	1e-173	89.37%	293	7OTE A
☒	DGY-06-116, a novel and selective covalent inhibitor of SRC kinase [Homo sapiens]	Homo sapiens	480	480	99%	1e-173	89.37%	276	6E6E A

E-value: This is the "Expect value", which is a measure of the number of matches expected to occur by chance in a database of the given size. A lower E-value indicates a more significant match.

Score: Raw score of the alignment, which is based on the number and quality of matching residues.

Query coverage: This is the percentage of the query sequence that was aligned with the subject sequence.

```
PyMOL>
/4mxo//A/257 261 266 271 276 281 286 291 296 301 306 311 316 321 326 331 336 341 346 351 356 361 366 371 376 381 386 391 396 401 406 411 424 431 436 441 446 451 456
KDAWEIPRESLRLEVKLGQGCFCGEVMMGTWNGTTRVAIKTLKPGTMSPEAFLEAQVMKKLRHEKLVQLYAVVSEEPYIVTEYMSKGSLLDFLKGETGKYLRLPQLVDMAAQIASGMAYVERMNYVHRDLRAANILVGENLVCKVADFGFLARLIE---FPIKWTAPAEALYGRFTIKSDVWSFGILLTELTTKGRV
/2oiq//A/256 261 266 271 276 281 286 291 296 301 306 311 316 321 326 331 336 341 346 351 356 361 366 371 376 381 386 391 396 401 406 421 426 431 436 441 446 451 456
AKDAWEIPRESLRLEVKLGQGCFCGEVMMGTWNGTTRVAIKTLKPGTMSPEAFLEAQVMKKLRHEKLVQLYAVVSEEPYIVTEYMSKGSLLDFLKGETGKYLRLPQLVDMAAQIASGMAYVERMNYVHRDLRAANILVGENLVCKVADFGFLARLIE---GAKFPIKWTAPAEALYGRFTIKSDVWSFGILLTELTTKGRV
/3u4w//A/259 266 271 276 281 286 291 296 301 306 311 316 321 326 331 336 341 346 351 356 361 366 371 376 381 386 391 396 401 406 411 416 421 426 431 436 441 446 451 456
AWEIPRESLRLEVKLGQGCFCGEVMMGTWNGTTRVAIKTLKPGTMSPEAFLEAQVMKKLRHEKLVQLYAVVSEEPYIVTEYMSKGSLLDFLKGETGKYLRLPQLVDMAAQIASGMAYVERMNYVHRDLRAANILVGENLVCKVADFGFLARLIEDNEYTARQGAKEPIKWTAPAEALYGRFTIKSDVWSFGILLTELTTKGRV
No License File - For Evaluation Only (0 days remaining)
```

- Compare possible templates. Go for structures with more crystallised residues
- Avoid the one with more missing residues

3U4W has no missing residues. Hence it is selected as template to model the structure.

Step 3 : Target-Template Alignment

- Query sequence aligned to the selected template sequence using 'align2d.py'.
- The alignment is written out in two formats. 'Target-template.ali' in PIR format and 'Target-template.pap' in PAP format.
- 'align2d.py' takes into account structural information from the template when constructing an alignment.
- PIR format is used by MODELLER in the subsequent model building stage, while the PAP alignment format is easier to inspect visually.
- In the PAP format, all identical positions are marked with a "*".

```
from modeller import *
env = Environ()
aln = Alignment(env)
mdl = Model(env, file='3u4w', model_segment=('FIRST:A','LAST:A'))
aln.append_model(mdl, align_codes='3u4wA',
atom_files='3u4w.pdb')
aln.append(file='Query.pir', align_codes='1YES')
aln.align2d(max_gap_length=50)
aln.write(file='TvLDH-1bdmA.ali', alignment_format='PIR')
aln.write(file='TvLDH-1bdmA.pap', alignment_format='PAP')
```

_aln.pos	10	20	30	40	50	60
1bdmA	MKAPVRVAVTGAAGQIGYSLLFRIAAGEMLGKDQPVILQLEIPQAMKALEGVVMELEDCAFPLLAGL					
TvLDH	MSEAAHVLITGAAGQIGYILSHWIASGELYG-DRQVYLHLLDIPPAMNRLTALTMELEDCAFPHLAGF					
_consrvd	* * *	*****	* ** *	* * *	* * *	* *****

_aln.p	70	80	90	100	110	120	130
1bdmA	EATDDPDVAFKADADYALLVGAAPRL-----QVNGKIFTEQGRALAEVAKKDVKVLVVGNPANTN						
TvLDH	VATTDPKAAFKDIDCAFLVASMPLKPGQVRADLISSNSVIFKNTGEYLSKWAKPSVKVLVIGNPDNTN						
_consrvd	** ** *	**** *	* ** *		* ** *	* ** *	*****

Step 4: Model Building

- After generating target-template alignment, we can move toward model building.
- For this, '*build.py*' script will be used. This script generates 5 models by default.
- Summary of all built models in the last section of the output log file.
- The log file gives a summary of all the models built. For each model, it lists the file name, which contains the coordinates of the model in PDB format.

>> Summary of successfully produced models:

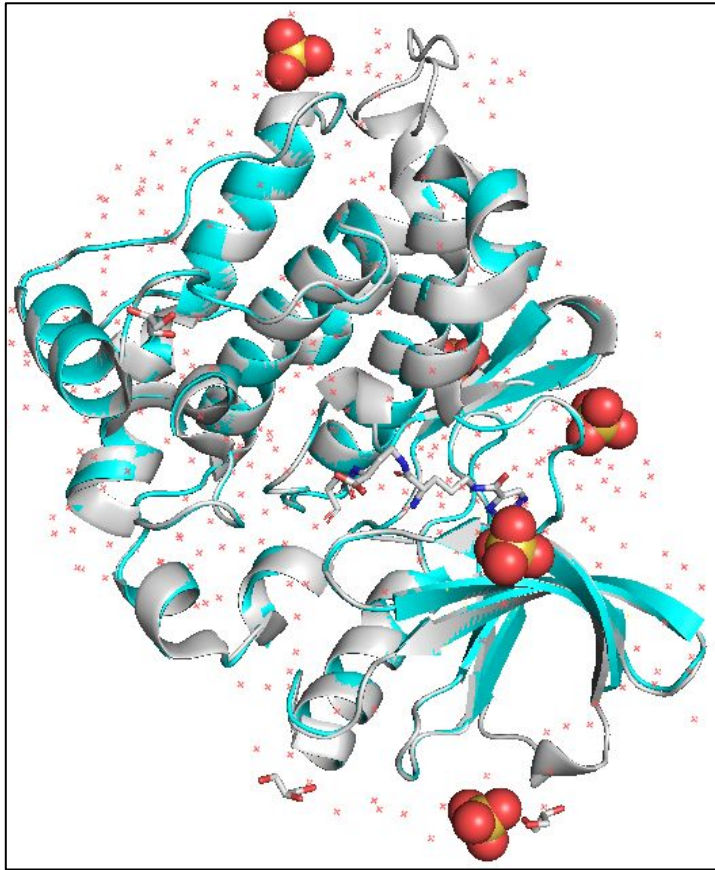
Filename	molpdf	DOPE score	GA341 score
1YES.B99990001.pdb	1166.22253	-31256.97266	1.00000
1YES.B99990002.pdb	1278.42578	-31383.46289	1.00000
1YES.B99990003.pdb	1119.32300	-31347.81836	1.00000
1YES.B99990004.pdb	1066.01270	-31371.45117	1.00000
1YES.B99990005.pdb	1115.09131	-31592.45898	1.00000

DOPE Score (Discrete Optimized Protein Energy)

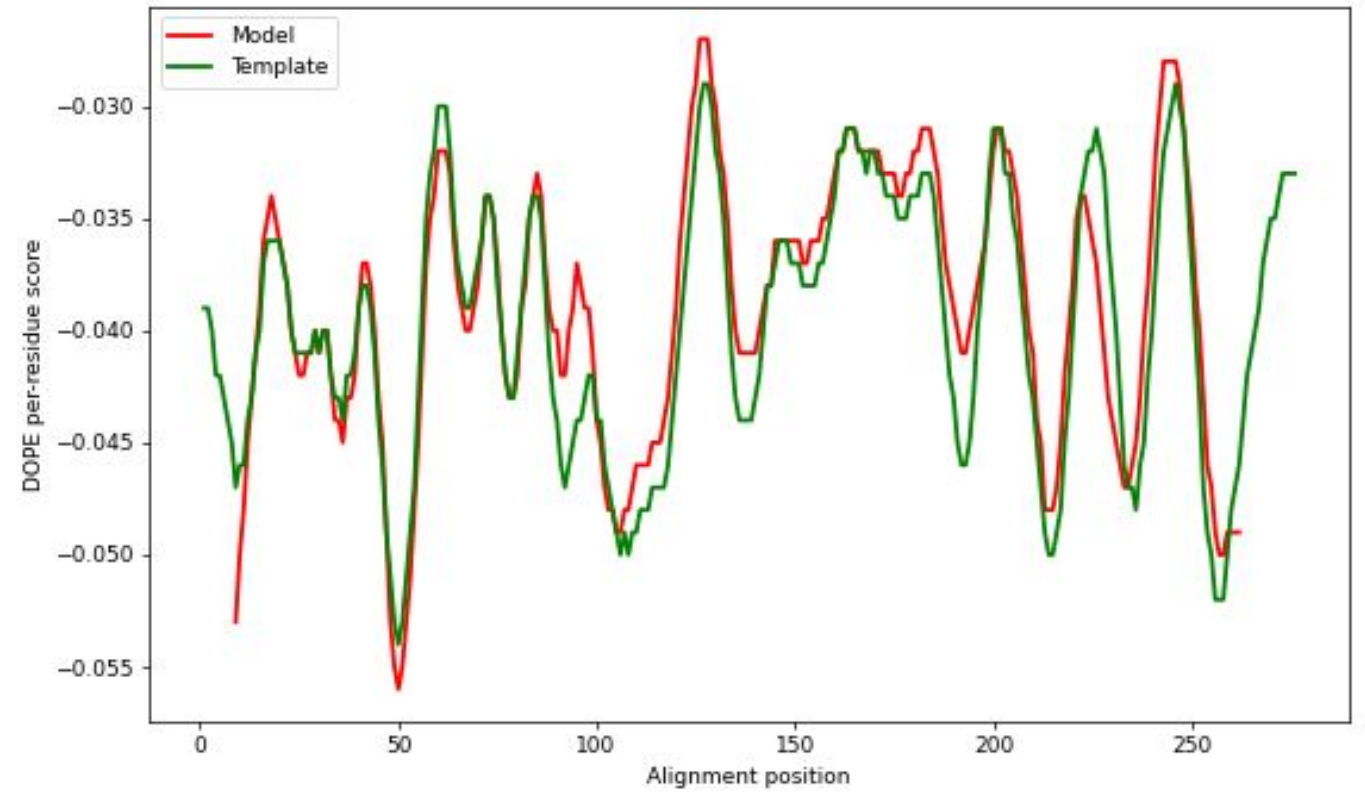
- Evaluates the quality of predicted structure
- Calculates the difference in energy of predicted and template structure based on physical and chemical principles (vdW, Hbonds, side-chain packing, etc.)
- Lower DOPE score ☞ More Similar model and template ☞ Good quality of model

Step 5: Model Evaluation

- Structural Alignment of Model with template using PyMol



RMSD-0.315



Model evaluated using DOPE score

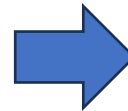
Modeling the structure of HCV NS3 protease (Advanced modeling)

STEP1. Download the query sequence & convert to .pir or .ali format

A. Visit ncbi page > set protein
database > search HCV NS3
Protease in the search box

B. Click on the first hit and
download the fasta sequence

The screenshot shows the NCBI Protein search results for the query 'hcv ns3 protease s'. The search results are displayed in a table with columns for Species, Source databases, Sequence length, Molecular weight, Release date, and Revision date. The first result is 'NS3 protease, partial [Hepacivirus hominis]' with 161 aa protein. The search results are sorted by Default order, and the page shows 1 to 20 of 21082 items. A blue arrow points from this page to the next screenshot.



The screenshot shows the NCBI Protein details page for the first hit, 'NS3 protease, partial [Hepacivirus hominis]'. The page displays the protein's GenBank accession number (AEO78507.1) and its FASTA sequence. A 'Choose Destination' dialog box is open, showing options to download the sequence as a File, Clipboard, Collections, or Analysis Tool. The 'File' option is selected, and the 'Format' is set to FASTA. The 'Create File' button is visible.

C. Convert and save downloaded fasta file to .ali/.pir format

.fasta



.ali

```
>AE078507.1 NS3 protease, partial [Hepacivirus hominis]  
LGCIIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVFHGAGSKTLAGPKGPITQMYTNVDKDLV  
GWQAPPGARSLTPCTCGGSDLYLVTRHADVIPVCRRGDSRGSLLSPRPVSYLKGSSGGPLLCPSGHAVGI  
FRAAVCTRGVAKAVDFIPVES
```



Align_code

```
>P1;NS3  
sequence:::::::::::::  
LGCIIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVFHGAGSKTLAGPKGPITQMYTNVDKDLV  
GWQAPPGARSLTPCTCGGSDLYLVTRHADVIPVCRRGDSRGSLLSPRPVSYLKGSSGGPLLCPSGHAVGI  
FRAAVCTRGVAKAVDFIPVES*
```

STEP 2. Run BLAST & search template

NIH National Library of Medicine
National Center for Biotechnology Information

Protein [Advanced](#) [Help](#)

FASTA Send to:

NS3 protease, partial [Hepacivirus hominis]
GenBank: AEO78507.1
[GenPept](#) [Identical Proteins](#) [Graphics](#)

>AEO78507.1 NS3 protease, partial [Hepacivirus hominis]
LGCIIITSLTGRDKNQVEGEVQVSTATQSFLATCVNGVCWTVFHGAGSKTLA
GPKGPITQMYTNVDKDLV
GWQAPPGARSLTPCTCGGSDLYLVTRHADVIPVCRRGDSRGSLLSPRPVSY
FRAAVCTRGVAKAVDFIPVES

Find regions of similarity between this sequence and other sequences using BLAST.

Related information
Nucleotide

OR

NIH National Library of Medicine
National Center for Biotechnology Information

BLAST® [Home](#) [Recent Results](#) [Saved Strategies](#) [Help](#)

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

Web BLAST

Nucleotide BLAST
nucleotide ► nucleotide

blastx
translated nucleotide ► protein

tblastn
protein ► translated nucleotide

Protein BLAST
protein ► protein

NEWS
Non-interactive searches of nt switch to core_nt
Starting late September 2024 all non-interactive WebBLAST and PrimerBLAST searches of "nt" will
Tue, 24 Sep 2024 [More BLAST news...](#)

From same ncbi page

From blast page

BLAST® » blastp suite

Standard Protein BLAST

BLASTP programs search protein databases using a protein query. [more...](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

>AEO78507.1 NS3 protease, partial [Hepacivirus hominis]
LGCIIITSLTGRDKNQVEGEVQVSTATQSFLATCVNGVCWTVFHGAGSKTLA
GPKGPITQMYTNVDKDLV
GWQAPPGARSLTPCTCGGSDLYLVTRHADVIPVCRRGDSRGSLLSPRPVSY

Query subrange [?](#)

From

To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

A. Runing BLAST

BLASTP programs search protein databases using a protein query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

>AEO78507.1 NS3 protease, partial [Hepacivirus hominis]
LGCIIITSLTGRDKNQVEGEVQVSTATQSFLATCVNGVCWTVFHGAGSKTLA
GPKGPIQMYTNVDKDLV
GWQAPPGARSLTPCTCGGSDLYLVTRHADVIPVCRRGDSRGSLLSPRPVSY

Query subrange [?](#)

From

To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

Standard

Database [?](#)

Organism [Optional](#)

Exclude [Optional](#)

Program Selection

Algorithm

- ☐ Non-redundant protein sequences (nr)
- ☐ RefSeq Select proteins (refseq_select)
- ☐ Reference proteins (refseq_protein)
- ☐ Model Organisms (landmark)
- ☐ UniProtKB/Swiss-Prot (swissprot)
- ☐ Patented protein sequences (pataa)
- ☒ Protein Data Bank proteins (pdb)
- ☐ Metagenomic proteins (env_nr)
- ☐ Transcriptome Shotgun Assembly proteins (tsa_nr)
- ☐ PHI-BLAST (Pattern Hit Initiated BLAST)
- ☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Program Selection

Algorithm

- ☐ Quick BLASTP (Accelerated protein-protein BLAST)
- ☒ blastp (protein-protein BLAST)
- ☐ PSI-BLAST (Position-Specific Iterated BLAST)
- ☐ PHI-BLAST (Pattern Hit Initiated BLAST)
- ☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

BLAST

Search database nr using Blastp (protein-protein BLAST)

☐ Show results in a new window

- Paste your fasta sequence or accession id or upload the fasta file in the query section



- Select database as Protein Data Bank proteins (pdb)



- In program selection select blastp as algorithm
- And hit BLAST

B. BLAST Results selection

Job Title	AE078507.1 NS3 protease, partial [Hepacivirus...
RID	U6SF6SV5013 Search expires on 02-06 20:04 pm Download All ▾
Program	BLASTP ? Citation ▾
Database	pdb See details ▾
Query ID	lcl Query_3240311
Description	AE078507.1 NS3 protease, partial [Hepacivirus hominis]
Molecule type	amino acid
Query Length	161
Other reports	Distance tree of results Multiple alignment MSA viewer ?

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[+ Add organism](#)

Percent Identity to **E value** to **Query Coverage** to

[Filter](#) [Reset](#)

Descriptions

Graphic Summary

Alignments

Taxonomy

Sequences producing significant alignments

DownloadSelect columnsShow100?

☒ select all80 sequences selected

GenPept

Graphics

Distance tree of results

Multiple alignment

MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ Chain A, NS3 PROTEASE [Hepatitis C virus (isolate BK)]	Hepatitis C virus (isolate BK)	321	321	100%	1e-113	96.89%	186	1NS3_A
☒ Chain A, NS3 SERINE PROTEASE [Hepacivirus hominis]	Hepacivirus hominis	320	320	100%	2e-113	96.89%	186	1BT7_A
☒ Chain A, NS3 protease [Hepacivirus hominis]	Hepacivirus hominis	320	320	100%	2e-113	96.89%	181	4K8B_A
☒ Chain A, NS3 PROTEINASE [Hepacivirus hominis]	Hepacivirus hominis	320	320	100%	3e-113	96.89%	189	1A1Q_A
☒ Chain A, NS3 protease [Hepatitis C virus (isolate 1)]	Hepatitis C virus (isolate 1)	320	320	100%	3e-113	96.27%	203	4TYD_A
☒ Chain A, Genome polyprotein [Hepacivirus hominis]	Hepacivirus hominis	320	320	100%	3e-113	96.89%	190	3KEE_A

- Select the PDBs with
1. High per. identity
 2. High query cover
 3. Low E-value

Selected PDBs
(for reference we
have selected
these 3 as they
had multiple
chains)

(here _A in PDB name specify the chain name)

STEP 3. download and align templates

A. Download the pdb 1NS3, 4K8B, 3KEE from pdb database

rcsb.org/structure/1NS3

RCSB PDB Deposit Search Visualize Analyze Download Learn About Careers COVID-19 Help Contact us MyPDB

RCSB PDB PROTEIN DATA BANK 231,029 Structures from the PDB 1,068,577 Computed Structure Models (CSM) Enter search term(s), Entry ID(s), Ligand ID or sequence Include CSM ? Search

Advanced Search | Browse Annotations Help

PDB-101 PDB EMDataResource NAKB wwPDB Foundation PDB-IHM

Structure Summary Structure Annotations Experiment Sequence Genome Versions

Biological Assembly 1 ?

1NS3

STRUCTURE OF HCV PROTEASE (BK STRAIN)

PDB DOI: <https://doi.org/10.2210/pdb1NS3/pdb>

Classification: **COMPLEX (HYDROLASE/PEPTIDE)**

Organism(s): Hepatitis C virus (isolate BK)

Expression System: Escherichia coli

Mutation(s): No

Deposited: 1997-04-05 Released: 1998-04-08

Deposition Author(s): Yan, Y., Munshi, S., Chen, Z.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 2.80 Å

R-Value Work: 0.204

R-Value Observed: 0.204

wwPDB Validation

Metric

Clashscore

Ramachandran outliers

Sidechain outliers

Worse

Percentile n

Percentile n

Display Files Download Files Data API

FASTA Sequence

PDBx/mmCIF Format

PDBx/mmCIF Format (gz)

BinaryCIF Format (gz)

PDB Format

PDB Format (gz)

PDBML/XML Format (gz)

Validation Full PDF

Validation (XML - gz)

Validation (CIF - gz)

Biological Assembly 1 (CIF - gz)

Biological Assembly 2 (CIF - gz)

Biological Assembly 3 (CIF - gz)

Biological Assembly 4 (CIF - gz)

Biological Assembly 1 (PDB - gz)

Biological Assembly 2 (PDB - gz)

report Full Report

Value

22

2.9%

15.4%

Better

Explore in 3D: Structure | Sequence Annotations | Validation Report | Ligand Interaction (ZN)

/download/1NS3.pdb

B. Align the templates

Alignment code

```
# Illustrates the SALIGN multiple structure/sequence alignment
```

```
from modeller import *
```

```
log.verbose()
```

```
env = Environ()
```

```
env.io.atom_files_directory = ['.', '../atom_files/']
```

```
aln = Alignment(env)
```

```
for (code, chain) in (('2mdh', 'A'), ('1bdm', 'A'), ('1b8p', 'A')):
```

```
    mdl = Model(env, file=code, model_segment=('FIRST:'+chain, 'LAST:'+chain))
```

```
    aln.append_model(mdl, atom_files=code, align_codes=code+chain)
```

```
for (weights, write_fit, whole) in (((1., 0., 0., 0., 1., 0.), False, True),  
                                     ((1., 0.5, 1., 1., 1., 0.), False, True),  
                                     ((1., 1., 1., 1., 1., 0.), True, False)):
```

```
    aln.salign(rms_cutoff=3.5, normalize_pp_scores=False,
```

```
               rr_file='$ (LIB)/as1.sim.mat', overhang=30,
```

```
               gap_penalties_1d=(-450, -50),
```

```
               gap_penalties_3d=(0, 3), gap_gap_score=0, gap_residue_score=0,
```

```
               dendrogram_file='fm00495.tree',
```

```
               alignment_type='tree', # If 'progressive', the tree is not
```

```
               # computed and all structures will be
```

```
               # aligned sequentially to the first
```

```
               feature_weights=weights, # For a multiple sequence alignment only
```

```
               # the first feature needs to be non-zero
```

```
               improve_alignment=True, fit=True, write_fit=write_fit,
```

```
               write_whole_pdb=whole, output='ALIGNMENT QUALITY')
```

```
aln.write(file='fm00495.pap', alignment_format='PAP')
```

```
aln.write(file='fm00495.ali', alignment_format='PIR')
```

```
aln.salign(rms_cutoff=1.0, normalize_pp_scores=False,
```

```
           rr_file='$ (LIB)/as1.sim.mat', overhang=30,
```

```
           gap_penalties_1d=(-450, -50), gap_penalties_3d=(0, 3),
```

```
           gap_gap_score=0, gap_residue_score=0, dendrogram_file='11s3A.tree',
```

```
           alignment_type='progressive', feature_weights=[0]*6,
```

```
           improve_alignment=False, fit=False, write_fit=True,
```

```
           write_whole_pdb=False, output='QUALITY')
```

change it with your template
pdb id and chain id

The alignment is then written
out in both PIR and PAP

formats

- Save this code as multiple_template_align.py file and run it in modeller prompt. It aligns all the template files.

Command: “**mod10.6**
multiple_template_align.py”

(where 10.6 is the modeller version)

Alignment result

```
_aln.pos      10      20      30      40      50      60
1ns3A      ITAYSQQTRGLLGCIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVYHGAGSKTLAGPKGP
4k8bA      ITAYSQQTRGLLGCIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVYHGAGSKTLAGPKGP
3keeA      ITAYSQQTRGLLGCIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVYHGAGSKTLAGPKGP
_consrvd     *****

_aln.p       70      80      90     100     110     120     130
1ns3A      ITQMYTNVDQDLVGWQAPPGARSLTPCTCGSSDLYLVTRHADVIPVRRRGDSRGSLLSPRPVSYLKGS
4k8bA      ITQMYTNVDQDLVGWQAPPGARSLTPCTCGSSDLYLVTRHADVIPVRRRGDSRGSLLSPRPVSYLKGS
3keeA      ITQMYTNVDQDLVGWQAPPGARSLTPCTCGSSDLYLVTRHADVIPVRRRGDSRGSLLSPRPVSYLKGS
_consrvd     *****

_aln.pos     140     150     160     170
1ns3A      SGGPLLCPSGHAVGIFRAAVCTRGVAKAVDFVPVESMETTMR
4k8bA      SGGPLLCPSGHAVGIFRAAVCTRGVAKAVDFVPVESMETTMR
3keeA      SGGPLLCPSGHAVGIFRAAVCTRGVAKAVDFVPVESMETTMR
_consrvd     *****
```

NOTE: this code can be found at
<https://salilab.org/modeller/tutorial/advanced.html>

STEP 4. Align the template with the query sequence

Alignment code

```
from modeller import *

log.verbose()
env = Environ()

env.libs.topology.read(file='${LIB}/top_heav.lib')

# Read aligned structure(s):
aln = Alignment(env)
aln.append(file='fm00495.ali', align_codes='all')
aln_block = len(aln)

# Read aligned sequence(s):
aln.append(file='TvLDH.ali', align_codes='TvLDH')

# Structure sensitive variable gap penalty sequence-sequence alignment:
aln.salign(output='', max_gap_length=20,
           gap_function=True, # to use structure-dependent gap penalty
           alignment_type='PAIRWISE', align_block=aln_block,
           feature_weights=(1., 0., 0., 0., 0., 0.), overhang=0,
           gap_penalties_1d=(-450, 0),
           gap_penalties_2d=(0.35, 1.2, 0.9, 1.2, 0.6, 8.6, 1.2, 0., 0.),
           similarity_flag=True)

aln.write(file='TvLDH-mult.ali', alignment_format='PIR')
aln.write(file='TvLDH-mult.pap', alignment_format='PAP')
```

- Save this code as template_query_align.py file and run it in modeller prompt

Command: “**mod10.6 template_query_align.py**”
(where 10.6 is the modeller version)

Alignment result

```
_aln.pos      10      20      30      40      50      60
1ns3A      ITAYSQQTRGLLGCIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVYHGAGSKTLAGPKGP
4k8bA      ITAYSQQTRGLLGCIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVYHGAGSKTLAGPKGP
3keeA      ITAYSQQTRGLLGCIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVYHGAGSKTLAGPKGP
NS3      -----LGCIIITSLTGRDKNQVEGEVQVVSTATQSFLATCVNGVCWTVFHGAGSKTLAGPKGP
_consrvd      *****

_aln.p      70      80      90      100     110     120     130
1ns3A      ITQMYTNVDQDLVGWQAPPGARSLTPCTCGSSDLYLVTRHADVIPVRRRGDSRGSLLSPRPVSYLKGS
4k8bA      ITQMYTNVDQDLVGWQAPPGARSLTPCTCGSSDLYLVTRHADVIPVRRRGDSRGSLLSPRPVSYLKGS
3keeA      ITQMYTNVDQDLVGWQAPPGARSLTPCTCGSSDLYLVTRHADVIPVRRRGDSRGSLLSPRPVSYLKGS
NS3      ITQMYTNVDKDLVGWQAPPGARSLTPCTCGSSDLYLVTRHADVIPVCRRGDSRGSLLSPRPVSYLKGS
_consrvd      *****

_aln.pos     140     150     160     170
1ns3A      SGGPLLCPSGHAVGIFRAAVCTRGVAKAVDFVPVESMETTMR
4k8bA      SGGPLLCPSGHAVGIFRAAVCTRGVAKAVDFVPVESMETTM-
3keeA      SGGPLLCPSGHAVGIFRAAVCTRGVAKAVDFVPVESMETTMR
NS3      SGGPLLCPSGHAVGIFRAAVCTRGVAKAVDFIPVES-----
_consrvd      *****
```

NOTE: this code can be found at
<https://salilab.org/modeller/tutorial/advanced.html>

STEP 5. model building

Code for model building

```
from modeller import *
from modeller.automodel import *

env = Environ()
a = AutoModel(env, alnfile='TvLDH-mult.ali',
               knowns=('1bdmA', '2mdhA', '1b8pA'), sequence='TvLDH')
a.starting_model = 1
a.ending_model = 5
a.make()
```

Change the name for this with the resultant .ali file
from the query-template alignment

Replace it with Align codes of your template
And sequence will be align code of the initial query .ali
file

- Save this code as build_model.py file and run it in modeller prompt
Command: “**mod10.6 build_model.py**”

- we build the new model for the NS3 target sequence based on the alignment against the multiple templates using the ‘build_model.py’ file:”

- We obtain 5 model of the provided query sequence and a log file with calculated enregies and molpdf score

```
>> Summary of successfully produced models:
Filename                                     molpdf
-----
NS3.B99990001.pdb                          5971.16797
NS3.B99990002.pdb                          5695.96875
NS3.B99990003.pdb                          5734.17041
NS3.B99990004.pdb                          5876.26660
NS3.B99990005.pdb                          5959.29102
```

- PDB with lowest molpdf score is selected.

A "molpdf score" refers to the "molecular probability density function" score, which is a standard metric used in the MODELLER software to evaluate the quality of a generated protein structure model; essentially, a lower molpdf score indicates a better fit to the known structural constraints and therefore a higher quality model.

STEP 6. Evaluate the model

Code for evaluating the model

```
from modeller import *
from modeller.scripts import complete_pdb

log.verbose() # request verbose output
env = Environ()
env.libs.topology.read(file='${LIB}/top_heav.lib') # read topology
env.libs.parameters.read(file='${LIB}/par.lib') # read parameters

# read model file
mdl = complete_pdb(env, 'TvLDH.B99990001.pdb')

# Assess all atoms with DOPE:
s = Selection(mdl)
s.assess_dope(output='ENERGY_PROFILE NO_REPORT', file='TvLDH.profile',
             normalize_profile=True, smoothing_window=15)
```

Replace it with resultant PDB file
name or PDB with lowest molpdf
score to evaluate the model

Rename the output
.profile file

- Save this code as evaluate_model.py file and run it in modeller prompt
Command: “**mod10.6 evaluate_model.py**”
- Finally, we use the DOPE potential to evaluate the new model coordinates using the ‘evaluate_model.py’ file

```
openf__224_> Open          1ns3.profile
# Energy of each residue is written to: 1ns3.profile
# The profile IS normalized by the number of restraints.
# The profiles are smoothed over a window of residues: 13
# The sum of all numbers in the file: -7.9059

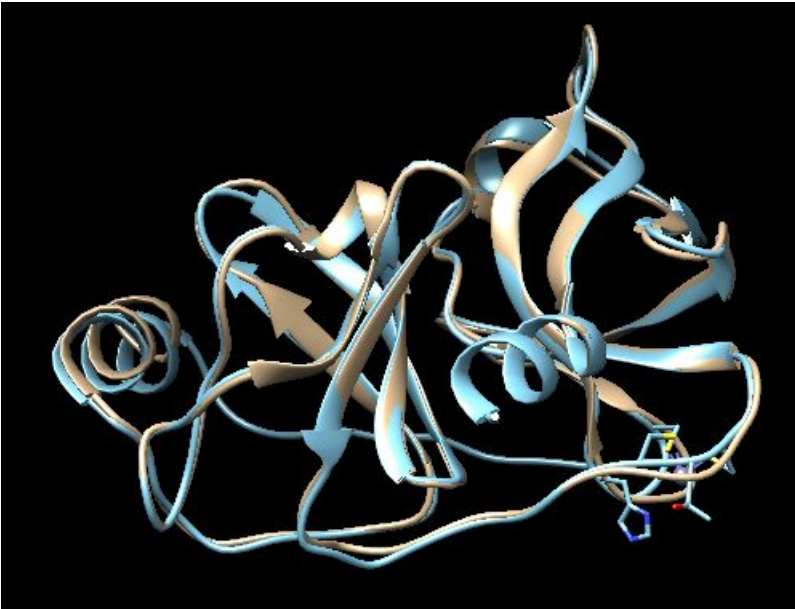
<< end of ENERGY.
DOPE score                  : -20308.244141

Dynamically allocated memory at      finish [B,KiB,MiB]:      8390465      8193.813
Starting time                    : 2025/01/31 15:01:20
Closing time                      : 2025/01/31 15:01:22
Total CPU time [seconds]          :          1.52
```

A .profile file and evaluate_model_profile.log (can find the dope score of the model) is obtained as an output. We can use .profile file to plot the dope score Vs Index plot.

STEP 7. Model visualisation using Chimera

- Open any template and modelled structure in chimera
- From the menu bar Select tools > structure analysis > match maker and align both the structure to visualise the alignment.



Matchmaker NS3.B99990003.pdb, chain A (#0) with 1ns3.pdb, chain A (#1), sequence alignment score = 773.7 with these parameters:
chain pairing: bb
Needleman-Wunsch using BLOSUM-62
ss fraction: 0.3
gap open (HH/SS/other) 18/18/6, extend 1
ss matrix: (O, S): -6 (H, O): -6 (H, H): 6 (S, S): 6 (H, S): -9 (O, O): 4
iteration cutoff: 2
RMSD between 159 pruned atom pairs is 0.515 angstroms; (across all 161 pairs: 0.571)

Reply log file with RMSD value (can be used to evaluate the model), can be obtained from favorites tab in the menu bar

STEP 8. evaluating the modelled structured using DOPE score profile plot

sourceforge.net/projects/gnuplot/files/gnuplot/

Business Software Open Source Software SourceForge Podcast Resources

Renderforest is an all-in-one branding platform
Customize videos with transitions, text, logo, and animation of your choice to promote and advance your social media presence.
Learn More

Home / Open Source Software / Multimedia / Graphics / gnuplot / Files

gnuplot Files

A portable, multi-platform, command-line driven graphing utility
Brought to you by: [broeker](#), [cgaylord](#), [hecking](#), [sfeam](#)

Summary Files Reviews Support Tickets Git Repository Mailing Lists Discussion News CVS Archive

Download Latest Version
gp602-win64-mingw.exe (49.2 MB) [Get Updates](#)

Home / gnuplot

Name	Modified	Size	Downloads / Week
Parent folder			
6.0.2	2024-12-23		3,317
testing	2024-12-06		0

- Download the gnuplot software from the provided link: <https://sourceforge.net/projects/gnuplot/files/gnuplot/>
- Complete the installation process as asked

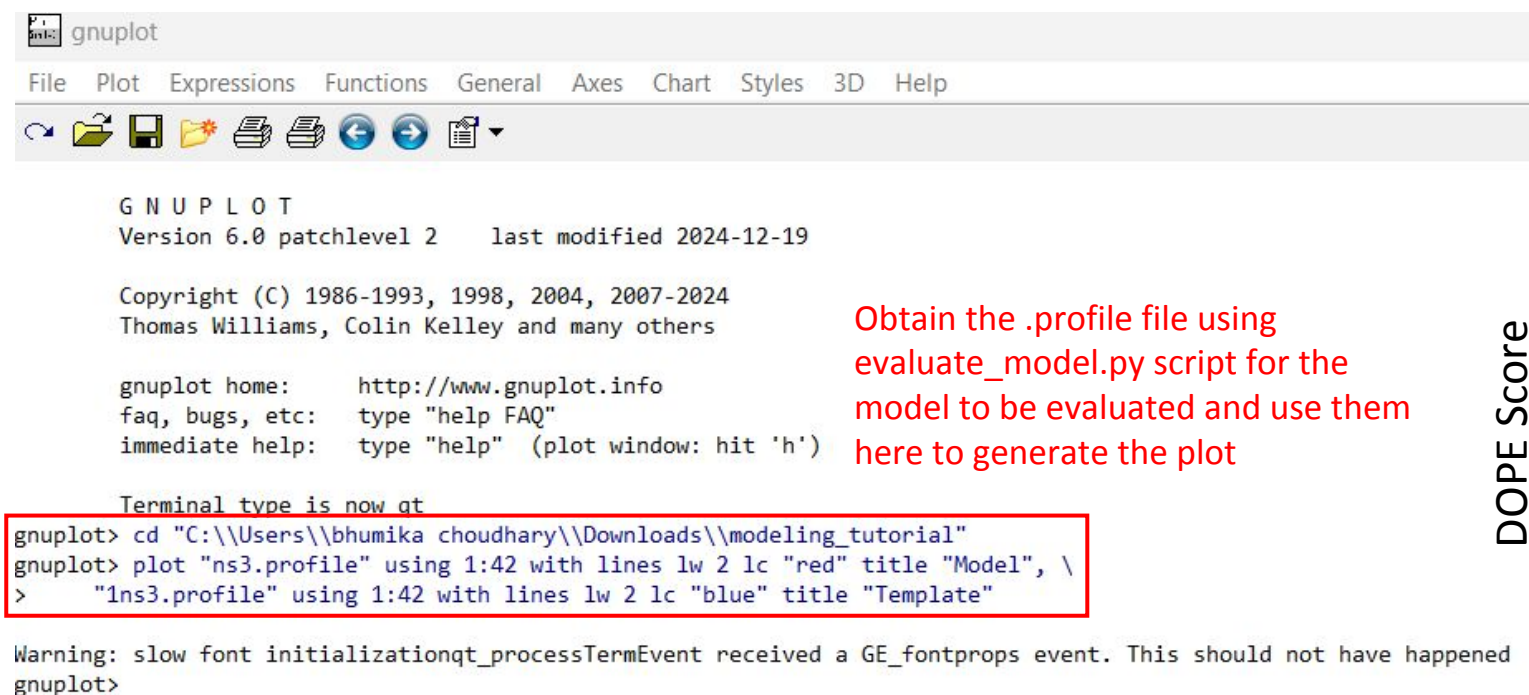
```
gnuplot
File Plot Expressions Functions General Axes Chart Styles 3D Help
G N U P L O T
Version 6.0 patchlevel 2    last modified 2024-12-19

Copyright (C) 1986-1993, 1998, 2004, 2007-2024
Thomas Williams, Colin Kelley and many others

gnuplot home:      http://www.gnuplot.info
faq, bugs, etc:    type "help FAQ"
immediate help:    type "help" (plot window: hit 'h')

Terminal type is now qt
gnuplot> _
```

gnuplot interface



gnuplot

File Plot Expressions Functions General Axes Chart Styles 3D Help

GNUPLOT
Version 6.0 patchlevel 2 last modified 2024-12-19

Copyright (C) 1986-1993, 1998, 2004, 2007-2024
Thomas Williams, Colin Kelley and many others

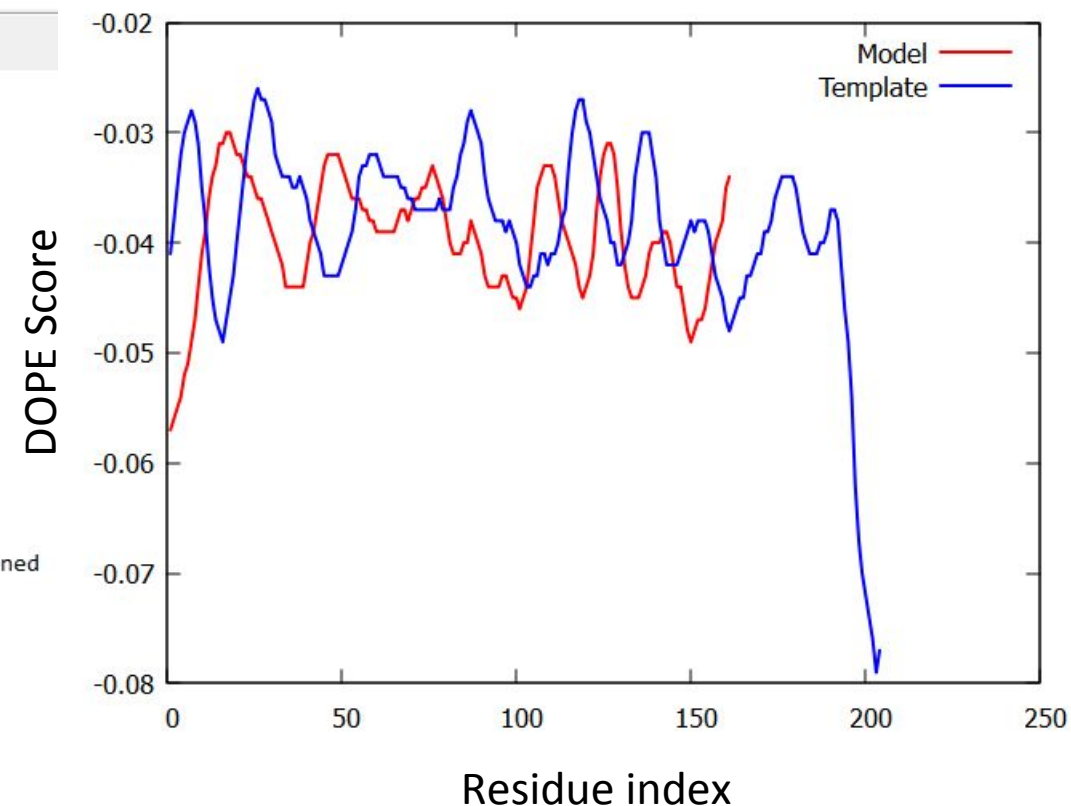
gnuplot home: http://www.gnuplot.info
faq, bugs, etc: type "help FAQ"
immediate help: type "help" (plot window: hit 'h')

Terminal type is now qt

```
gnuplot> cd "C:\\Users\\bhumika choudhary\\Downloads\\modeling_tutorial"  
gnuplot> plot "ns3.profile" using 1:42 with lines lw 2 lc "red" title "Model", \  
> "1ns3.profile" using 1:42 with lines lw 2 lc "blue" title "Template"
```

Warning: slow font initializationqt_processTermEvent received a GE_fontprops event. This should not have happened
gnuplot>

Obtain the .profile file using
evaluate_model.py script for the
model to be evaluated and use them
here to generate the plot



- Set the current working directory as shown in the above figure
- Type the following commands and shown to plot the dope score Vs residue index plot
- You can also use matplotlib python package to obtain this plot using .profile files of the models